Mode 6 — Square-BFP (centered / opt)

DP32 (8×8) → DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → block exponents → DP8 (4×4) → centered aligned add-and-square

Step 1 — DP32 (8×8)

$$\mathrm{DP32}\,(8\times 8) = \sum_{i=0}^{31} a_i\, b_i$$

Step 2 — DP16 (8×8)

$$\mathrm{DP32}\,(8\times 8) = \underbrace{\sum_{i=0}^{15} a_i\, b_i}_{\mathrm{DP16}\,(8\times 8)} + \underbrace{\sum_{i=16}^{31} a_i\, b_i}_{\mathrm{DP16}\,(8\times 8)}$$

Step 3 — DP16 (8×4)

$$\mathrm{DP32}\,(8\times 8) = 2^4 \underbrace{\sum_{i=0}^{15} a_i\, b_i^{hi}}_{\mathrm{DP16}\,(8\times 4)} + \underbrace{\sum_{i=0}^{15} a_i\, b_i^{lo}}_{\mathrm{DP16}\,(8\times 4)} + 2^4 \underbrace{\sum_{i=16}^{31} a_i\, b_i^{hi}}_{\mathrm{DP16}\,(8\times 4)} + \underbrace{\sum_{i=16}^{31} a_i\, b_i^{lo}}_{\mathrm{DP16}\,(8\times 4)}$$

Step 4 — DP8 (8×4)

$$\begin{aligned}\mathrm{DP32}\,(8\times 8) = {} & 2^4 \underbrace{\sum_{i=0}^{7} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times 4)} + \underbrace{\sum_{i=0}^{7} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times 4)} + \underbrace{\sum_{i=8}^{15} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times 4)} \\ & + 2^4 \underbrace{\sum_{i=16}^{23} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times 4)} + 2^4 \underbrace{\sum_{i=24}^{31} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(8\times 4)} + \underbrace{\sum_{i=16}^{23} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times 4)} + \underbrace{\sum_{i=24}^{31} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(8\times 4)}\end{aligned}$$

Step 5 — Block exponents (one exponent block per DP8 (8×4))

$$\left(2^4\sum_{i=8q}^{8q+7} a_i b_i^{hi} + \sum_{i=8q}^{8q+7} a_i b_i^{lo},\ E_q\right)_{q=0..3} \longrightarrow \left(\ M = \sum_{q=0}^{3} 2^{-\delta_q}\left(2^4\sum_{i=8q}^{8q+7} a_i b_i^{hi} + \sum_{i=8q}^{8q+7} a_i b_i^{lo}\right),\ \ E\ \right)$$

$E_q = E_{a,q} + E_{b,q}$; $E = \max(E_0, \ldots, E_3)$, $\delta_q = E - E_q$ (staged, telescoping); truncating shifts, on block sums only; $\mathrm{DP32}\,(8\times 8) = M \cdot 2^E$.

Step 6 — DP8 (4×4)

$$\begin{aligned}M = {} & 2^{-\delta_0}\Big(2^8 \underbrace{\sum_{i=0}^{7} a_{H,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times 4)} + 2^4 \underbrace{\sum_{i=0}^{7} a_{L,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times 4)} + 2^4 \underbrace{\sum_{i=0}^{7} a_{H,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times 4)} + \underbrace{\sum_{i=0}^{7} a_{L,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times 4)}\Big) \\ & + 2^{-\delta_1}\Big(2^8 \underbrace{\sum_{i=8}^{15} a_{H,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{L,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{H,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times 4)} + \underbrace{\sum_{i=8}^{15} a_{L,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times 4)}\Big) \\ & + 2^{-\delta_2}\Big(2^8 \underbrace{\sum_{i=16}^{23} a_{H,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times 4)} + 2^4 \underbrace{\sum_{i=16}^{23} a_{L,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times 4)} + 2^4 \underbrace{\sum_{i=16}^{23} a_{H,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times 4)} + \underbrace{\sum_{i=16}^{23} a_{L,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times 4)}\Big) \\ & + 2^{-\delta_3}\Big(2^8 \underbrace{\sum_{i=24}^{31} a_{H,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times 4)} + 2^4 \underbrace{\sum_{i=24}^{31} a_{L,i}\, b_i^{hi}}_{\mathrm{DP8}\,(4\times 4)} + 2^4 \underbrace{\sum_{i=24}^{31} a_{H,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times 4)} + \underbrace{\sum_{i=24}^{31} a_{L,i}\, b_i^{lo}}_{\mathrm{DP8}\,(4\times 4)}\Big)\end{aligned}$$

Step 7 — Centered aligned add-and-square (subtract 8 from unsigned nibbles a_L, b^{lo}; per DP8 (4×4) add $c = 0/512/2048$ for 0/1/2 unsigned nibbles; four rows per quarter q, each weighted $2^{-\delta_q}$)

$$
\begin{aligned}
M = {} & 2^{-\delta_0}\, 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a_{H,i}+b_i^{hi})^2 - \sum_{i=0}^{7} a_{H,i}^2 - \sum_{i=0}^{7}(b_i^{hi})^2\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_0}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a_{L,i}-8)+b_i^{hi}\big)^2 - \sum_{i=0}^{7}(a_{L,i}-8)^2 - \sum_{i=0}^{7}(b_i^{hi}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_0}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big(a_{H,i}+(b_i^{lo}-8)\big)^2 - \sum_{i=0}^{7}(a_{H,i}-8)^2 - \sum_{i=0}^{7}(b_i^{lo}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_0}\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2 - \sum_{i=0}^{7}(a_{L,i}-16)^2 - \sum_{i=0}^{7}(b_i^{lo}-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_1}\, 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}(a_{H,i}+b_i^{hi})^2 - \sum_{i=8}^{15} a_{H,i}^2 - \sum_{i=8}^{15}(b_i^{hi})^2\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_1}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}\big((a_{L,i}-8)+b_i^{hi}\big)^2 - \sum_{i=8}^{15}(a_{L,i}-8)^2 - \sum_{i=8}^{15}(b_i^{hi}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_1}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}\big(a_{H,i}+(b_i^{lo}-8)\big)^2 - \sum_{i=8}^{15}(a_{H,i}-8)^2 - \sum_{i=8}^{15}(b_i^{lo}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_1}\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2 - \sum_{i=8}^{15}(a_{L,i}-16)^2 - \sum_{i=8}^{15}(b_i^{lo}-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_2}\, 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=16}^{23}(a_{H,i}+b_i^{hi})^2 - \sum_{i=16}^{23} a_{H,i}^2 - \sum_{i=16}^{23}(b_i^{hi})^2\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_2}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=16}^{23}\big((a_{L,i}-8)+b_i^{hi}\big)^2 - \sum_{i=16}^{23}(a_{L,i}-8)^2 - \sum_{i=16}^{23}(b_i^{hi}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_2}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=16}^{23}\big(a_{H,i}+(b_i^{lo}-8)\big)^2 - \sum_{i=16}^{23}(a_{H,i}-8)^2 - \sum_{i=16}^{23}(b_i^{lo}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_2}\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=16}^{23}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2 - \sum_{i=16}^{23}(a_{L,i}-16)^2 - \sum_{i=16}^{23}(b_i^{lo}-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_3}\, 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=24}^{31}(a_{H,i}+b_i^{hi})^2 - \sum_{i=24}^{31} a_{H,i}^2 - \sum_{i=24}^{31}(b_i^{hi})^2\Big)}_{\text{DP8 (4×4)}} \\
& + 2^{-\delta_3}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=24}^{31}\big((a_{L,i}-8)+b_i^{hi}\big)^2 - \sum_{i=24}^{31}(a_{L,i}-8)^2 - \sum_{i=24}^{31}(b_i^{hi}-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}
\end{aligned}
$$

$$+\,2^{-\delta_3}\,2^4\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=24}^{31}\big(a_{H,i}+(b_i^{lo}-8)\big)^2-\sum_{i=24}^{31}(a_{H,i}-8)^2-\sum_{i=24}^{31}(b_i^{lo}-8)^2+512\Big)}_{\text{DP8 }(4\times4)}$$

$$+\,2^{-\delta_3}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=24}^{31}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2-\sum_{i=24}^{31}(a_{L,i}-16)^2-\sum_{i=24}^{31}(b_i^{lo}-16)^2+2048\Big)}_{\text{DP8 }(4\times4)}$$

Step 8 — All components

$$\boxed{\begin{aligned}
\mathrm{PE} = {}& 2^{-\delta_0}\Big(2^8\sum_{i=0}^{7}(a_{H,i}+b_i^{hi})^2+2^4\sum_{i=0}^{7}\big((a_{L,i}-8)+b_i^{hi}\big)^2+2^4\sum_{i=0}^{7}\big(a_{H,i}+(b_i^{lo}-8)\big)^2+\sum_{i=0}^{7}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2\Big)\\
&+2^{-\delta_1}\Big(2^8\sum_{i=8}^{15}(a_{H,i}+b_i^{hi})^2+2^4\sum_{i=8}^{15}\big((a_{L,i}-8)+b_i^{hi}\big)^2+2^4\sum_{i=8}^{15}\big(a_{H,i}+(b_i^{lo}-8)\big)^2+\sum_{i=8}^{15}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2\Big)\\
&+2^{-\delta_2}\Big(2^8\sum_{i=16}^{23}(a_{H,i}+b_i^{hi})^2+2^4\sum_{i=16}^{23}\big((a_{L,i}-8)+b_i^{hi}\big)^2+2^4\sum_{i=16}^{23}\big(a_{H,i}+(b_i^{lo}-8)\big)^2+\sum_{i=16}^{23}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2\Big)\\
&+2^{-\delta_3}\Big(2^8\sum_{i=24}^{31}(a_{H,i}+b_i^{hi})^2+2^4\sum_{i=24}^{31}\big((a_{L,i}-8)+b_i^{hi}\big)^2+2^4\sum_{i=24}^{31}\big(a_{H,i}+(b_i^{lo}-8)\big)^2+\sum_{i=24}^{31}\big((a_{L,i}-8)+(b_i^{lo}-8)\big)^2\Big)\\
\alpha = {}& 2^{-\delta_0}\Big(2^8\sum_{i=0}^{7}a_{H,i}^2+2^4\sum_{i=0}^{7}(a_{L,i}-8)^2+2^4\sum_{i=0}^{7}(a_{H,i}-8)^2+\sum_{i=0}^{7}(a_{L,i}-16)^2\Big)\\
&+2^{-\delta_1}\Big(2^8\sum_{i=8}^{15}a_{H,i}^2+2^4\sum_{i=8}^{15}(a_{L,i}-8)^2+2^4\sum_{i=8}^{15}(a_{H,i}-8)^2+\sum_{i=8}^{15}(a_{L,i}-16)^2\Big)\\
&+2^{-\delta_2}\Big(2^8\sum_{i=16}^{23}a_{H,i}^2+2^4\sum_{i=16}^{23}(a_{L,i}-8)^2+2^4\sum_{i=16}^{23}(a_{H,i}-8)^2+\sum_{i=16}^{23}(a_{L,i}-16)^2\Big)\\
&+2^{-\delta_3}\Big(2^8\sum_{i=24}^{31}a_{H,i}^2+2^4\sum_{i=24}^{31}(a_{L,i}-8)^2+2^4\sum_{i=24}^{31}(a_{H,i}-8)^2+\sum_{i=24}^{31}(a_{L,i}-16)^2\Big)\\
\beta = {}& 2^{-\delta_0}\Big(2^8\sum_{i=0}^{7}(b_i^{hi})^2+2^4\sum_{i=0}^{7}(b_i^{hi}-8)^2+2^4\sum_{i=0}^{7}(b_i^{lo}-8)^2+\sum_{i=0}^{7}(b_i^{lo}-16)^2\Big)\\
&+2^{-\delta_1}\Big(2^8\sum_{i=8}^{15}(b_i^{hi})^2+2^4\sum_{i=8}^{15}(b_i^{hi}-8)^2+2^4\sum_{i=8}^{15}(b_i^{lo}-8)^2+\sum_{i=8}^{15}(b_i^{lo}-16)^2\Big)\\
&+2^{-\delta_2}\Big(2^8\sum_{i=16}^{23}(b_i^{hi})^2+2^4\sum_{i=16}^{23}(b_i^{hi}-8)^2+2^4\sum_{i=16}^{23}(b_i^{lo}-8)^2+\sum_{i=16}^{23}(b_i^{lo}-16)^2\Big)\\
&+2^{-\delta_3}\Big(2^8\sum_{i=24}^{31}(b_i^{hi})^2+2^4\sum_{i=24}^{31}(b_i^{hi}-8)^2+2^4\sum_{i=24}^{31}(b_i^{lo}-8)^2+\sum_{i=24}^{31}(b_i^{lo}-16)^2\Big)\\
C = {}& \big(2^{-\delta_0}+2^{-\delta_1}+2^{-\delta_2}+2^{-\delta_3}\big)\big(0+8192+8192+2048\big)=\big(2^{-\delta_0}+2^{-\delta_1}+2^{-\delta_2}+2^{-\delta_3}\big)\cdot 18432\\
M = {}& \tfrac{1}{2}\big(\mathrm{PE}-\alpha-\beta+C\big),\qquad E=\max(E_0,E_1,E_2,E_3)
\end{aligned}}$$